In [1]:
```
import tensorflow as  tf
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D,MaxPooling2D,Flatten,Dense
from tensorflow.keras.preprocessing.image import ImageDataGenerator
import matplotlib.pyplot as plt
import os
```

In [2]:
```
base_dir=r"C:\Users\cs2\Desktop\sadhika"
img_size=124
```

In [3]:
```
datagen=ImageDataGenerator(
rescale=1./255,
validation_split=0.2,
shear_range=0.2,
zoom_range=0.2,
horizontal_flip=True
)
```

In [4]:
```
train_gen=datagen.flow_from_directory(
base_dir,
target_size=(img_size,img_size),batch_size=32,
class_mode='binary',
subset='training',
shuffle=True
)
```

```
Found 800 images belonging to 3 classes.
```

In [10]:
```
val_gen=datagen.flow_from_directory(
base_dir,
target_size=(img_size,img_size),
batch_size=32,
class_mode='binary',
subset='validation',
shuffle=True
)
```

```
Found 199 images belonging to 3 classes.
```

In [11]:
```
model=Sequential([
    Conv2D(6,kernel_size=(5,5),activation='relu',input_shape=(img_size,img_size,
    MaxPooling2D(pool_size=(2,2)),
    Conv2D(10,kernel_size=(5,5),activation='relu'),
    MaxPooling2D(pool_size=(2,2)),
    Flatten(),
    Dense(120,activation='relu'),
    Dense(84,activation='relu'),
    Dense(1,activation='sigmoid')

])
```

```
C:\Users\cs2\AppData\Roaming\Python\Python313\site-packages\keras\src\layers\conv
olutional\base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an `Input(shape)`
object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

In [12]:
```
model.compile(optimizer='adam',loss='binary_crossentropy',metrics=['accuracy'])
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Pleas
e use WSL2 or the TensorFlow-DirectML plugin.
```

In [13]:
```
history=model.fit(train_gen,validation_data=val_gen,epochs=5)
```

```
Epoch 1/5
25/25 ━━━━━━━━━━━━━━━━━━━━ 11s 397ms/step - accuracy: 0.5000 - loss: -504.7106 -
val_accuracy: 0.4975 - val_loss: -3013.7544
Epoch 2/5
25/25 ━━━━━━━━━━━━━━━━━━━━ 4s 163ms/step - accuracy: 0.5000 - loss: -30634.6992 -
val_accuracy: 0.4975 - val_loss: -112386.5391
Epoch 3/5
25/25 ━━━━━━━━━━━━━━━━━━━━ 4s 164ms/step - accuracy: 0.5000 - loss: -510298.6875
- val_accuracy: 0.4975 - val_loss: -1276062.2500
Epoch 4/5
25/25 ━━━━━━━━━━━━━━━━━━━━ 4s 175ms/step - accuracy: 0.5000 - loss: -3482255.0000
- val_accuracy: 0.4975 - val_loss: -7636917.5000
Epoch 5/5
25/25 ━━━━━━━━━━━━━━━━━━━━ 4s 162ms/step - accuracy: 0.5000 - loss: -17075002.000
0 - val_accuracy: 0.4975 - val_loss: -30495620.0000
```

In [14]:
```
loss,acc=model.evaluate(val_gen)
print(f"Validation Accuracy:{acc*100:.2f}%")
```

```
7/7 ━━━━━━━━━━━━━━━━━━━━ 1s 107ms/step - accuracy: 0.4975 - loss: -30643982.0000
Validation Accuracy:49.75%
```

In [15]:
```
x_batchy_batch = next(val_gen)
preds = model.predict(x_batchy_batch)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 96ms/step
C:\Users\cs2\AppData\Roaming\Python\Python313\site-packages\keras\src\models\func
tional.py:259: UserWarning: The structure of `inputs` doesn't match the expected
structure.
Expected: keras_tensor
Received: inputs=('Tensor(shape=(32, 124, 124, 3))', 'Tensor(shape=(32,))')
  warnings.warn(msg)
```

In [16]:
```
x_batch, y_batch = next(val_gen)
preds = model.predict(x_batch)

plt.figure(figsize=(15, 6))

for i in range(10):
    plt.subplot(2, 5, i + 1)

    plt.imshow(x_batch[i].reshape(img_size, img_size, 3))

    pred_label = 'Dog' if preds[i] > 0.5 else 'Cat'
    actual_label = 'Dog' if y_batch[i] == 1 else 'Cat'

    plt.title(f"Pred: {pred_label}\nActual: {actual_label}")
    plt.axis('off')

plt.tight_layout()
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step
```


Pred: Dog
Actual: Cat
Pred: Dog
Actual: Dog
Pred: Dog
Actual: Dog
Pred: Dog
Actual: Cat
Pred: Dog
Actual: Cat
Pred: Dog
Actual: Cat
Pred: Dog
Actual: Dog
Pred: Dog
Actual: Dog
Pred: Dog
Actual: Dog
Pred: Dog
Actual: Dog

In [ ]: